

TUOLE, China

WMO: 526330

Lat: 38.8087N

Lon: 98.4179E

Elev: 11056

StdP: 9.70

Time Zone: 8.00 (E08)

Period: 82-98

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -20.1	(c) -16.0	(d) -36.8	(e) 1.0	(f) -5.0	(g) -34.2	(h) 1.2	(i) -2.1	(j) 33.0	(k) 14.4	(l) 29.2	(m) 19.4	(n) 0.4	(o) N/A	(p) N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 19.6	(c) 68.6	(d) 48.3	(e) 65.5	(f) 46.8	(g) 62.7	(h) 45.5	(i) 51.8	(j) 62.5	(k) 49.6	(l) 60.3	(m) 48.0	(n) 58.4	(o) 6.0	(p) N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
48.3	76.3	54.5	45.7	69.0	52.9	43.5	63.4	51.5	25.9	62.8	24.3	60.4	23.2	58.5	

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 28.6	(b) 23.1	(c) 19.4	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A		
(a) 28.6	(b) 23.1	(c) 19.4	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	28.6	2.4	9.4	18.7	29.4	41.1	46.9	51.3	49.5	41.4	29.5	16.3	6.6
	DBStd	17.79	6.44	7.08	6.82	5.91	5.87	4.95	4.21	5.09	4.28	5.49	6.03	5.98
	HDD50	7954	1476	1138	969	617	282	117	31	71	262	636	1011	1344
	HDD65	13270	1941	1558	1434	1067	741	542	425	482	709	1101	1461	1809
	CDD50	158	0	0	0	0	5	24	72	54	3	0	0	0
	CDD65	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDH74	4	0	0	0	0	0	0	1	3	0	0	0	0
	CDH80	0	0	0	0	0	0	0	0	0	0	0	0	0
(14) Wind	WSAvg	5.1	3.2	4.6	6.9	7.6	6.9	6.0	5.8	5.2	4.6	3.8	3.6	2.8
(15) Precipitation	PrecAvg	11.2	0.0	0.1	0.2	0.4	1.4	2.4	3.0	2.3	1.1	0.3	0.0	0.0
	PrecMax	15.0	0.2	0.4	0.5	1.1	3.4	5.1	5.2	3.9	2.9	0.7	0.2	0.1
	PrecMin	8.0	0.0	0.0	0.0	0.0	0.0	0.9	1.5	0.9	0.0	0.0	0.0	0.0
	PrecStd	1.8	0.0	0.1	0.1	0.3	0.7	1.0	0.9	0.8	0.7	0.2	0.1	0.0
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	29.6	40.2	46.3	58.9	65.7	67.9	73.7	72.9	65.4	54.6	42.6	31.7
		MCWB	19.5	25.8	28.9	36.3	42.3	45.3	52.4	49.4	49.6	35.3	28.3	21.2
	2%	DB	24.0	33.8	42.0	52.6	61.3	65.0	69.1	68.6	59.4	49.7	37.8	27.1
		MCWB	15.8	22.4	27.0	32.9	40.6	44.4	49.9	48.0	42.1	32.8	25.4	17.9
	5%	DB	20.4	29.3	38.2	47.8	57.3	62.0	65.5	65.0	56.2	46.1	34.3	23.9
		MCWB	13.2	19.6	25.4	30.6	38.3	43.5	48.1	47.1	40.1	31.5	23.0	16.0
	10%	DB	17.0	25.6	34.3	43.9	53.8	59.0	62.1	61.8	53.4	42.8	30.8	20.5
		MCWB	10.8	17.1	23.4	29.0	36.6	42.5	46.5	46.5	38.7	29.6	21.2	13.6
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	19.8	26.2	29.8	37.2	44.7	49.5	54.9	55.2	52.9	36.5	28.9	21.4
		MCDB	29.1	39.3	42.3	56.6	62.9	60.9	68.7	61.9	60.5	50.9	41.5	30.9
	2%	WB	15.9	22.5	27.9	33.8	41.9	47.4	51.7	52.0	45.7	34.0	25.7	18.5
		MCDB	23.4	33.2	40.4	49.8	56.6	59.1	63.5	61.5	53.7	47.6	37.0	26.8
	5%	WB	13.2	19.9	26.0	31.5	40.2	45.7	49.7	49.8	42.8	32.1	23.5	16.1
		MCDB	20.0	29.0	37.5	45.8	53.7	57.6	60.7	59.9	52.0	43.0	33.7	23.6
	10%	WB	10.7	17.3	23.8	29.5	38.4	44.1	48.2	48.0	40.7	30.6	21.3	13.7
		MCDB	16.7	25.4	33.8	42.1	50.5	55.1	59.2	57.9	50.1	40.9	30.6	20.2
(35) Mean Daily Temperature Range	5% DB	MDBR	30.9	31.2	29.5	24.4	23.4	20.3	19.6	21.7	25.5	27.6	30.6	30.7
		MCDBR	34.5	34.9	34.3	31.5	28.0	26.2	25.8	28.0	30.2	31.2	33.9	32.5
		MCWBR	27.5	25.9	22.8	15.7	12.0	10.6	10.5	10.8	14.3	15.6	22.7	24.8
	5% WB	MCDBR	34.0	33.7	32.1	27.0	23.5	20.7	21.5	19.6	23.7	27.5	32.4	31.8
		MCWBR	27.3	24.9	21.2	14.5	10.8	9.2	9.8	9.4	12.6	14.5	21.6	24.6
(40) Clear-Sky Solar Irradiance	taub		0.223	0.262	0.323	0.383	0.407	0.406	0.386	0.374	0.335	0.289	0.242	0.215
	taud		2.453	2.327	2.116	1.969	1.976	2.069	2.218	2.242	2.310	2.343	2.436	2.488
	Ebn at Noon		312	308	295	282	277	277	281	282	288	293	301	309
	Edh at Noon		25	33	45	56	57	52	44	42	37	32	25	22
(44) All-Sky Solar Radiation	RadAvg		901	1169	1564	1871	1996	1993	1939	1812	1563	1318	993	831
	RadStd		52	67	99	102	108	132	124	124	95	69	53	46

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (3 neighbors)	+0.50	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+178	+81

Nomenclature: See separate page